Description of the data (Yash)

For our project we will be using the Cassava Leaf Disease Classification dataset from Kaggle as this dataset has/contains 21,367 images of cassava leaves that were collected from the fields in Uganda and each of the image is labeled as either a healthy leaf or one of the four different cassava diseases. Our goal is to use these images to train and compare the different machine learning models that can identify the disease from a picture of a cassava leaf. One of the main challenges with this dataset is that the images are not equally distributed among all the five classes and for example, Cassava Mosaic Disease (CMD) makes up approximately 61 percent of the dataset and while the rarest disease class only makes up about 5 percent of the dataset. This creates a problem because a machine learning model will become better at predicting the majority class while having difficulty identifying the diseases that have fewer/lesser images. This is important for our project because we are not only trying to achieve/get high overall accuracy but we also want our models to be able to identify the rare diseases correctly as to address this problem we plan to compare the different approaches for handling the imbalanced in the dataset, focal loss, including oversampling, and class balanced loss. We will use the images as a input for our machine learning models and then comparing their performance using accuracy, recall for each disease, confusion matrices, and macro f1 score as this will help us understand how well each model performs across all the five classes rather than only focusing on the overall accuracy.

What remains to be done (Yash)